

Minimal Disruption-Reconfiguration problem over WDM Optical Networks

A new approach based on Multi-objective Ant Colony optimization

Enrique Dávalos, Benjamín Barán, Diego Pinto

Polytechnical School

National University of Asunción

San Lorenzo, Paraguay

{edavalos, bbaran, dpinto}@pol.una.py

Abstract—Management of dynamic traffic in WDM optical network is a critical problem in Optical Telecommunications. The need of minimizing the utilization of network capacity is present in several phases of network operation. Re-routing the current working paths to reach good performance of dynamic assigned resources is a crucial task. On the other hand, the quality-of-service (QoS) is worse when disruptions are generated by re-routing. Therefore, re-routing and QoS are functions in conflict and must be treated as a multi-objective optimization problem to accomplish the best decision. This work proposes a multi-objective optimization problem approach based on a Multi-objective Ant Colony Optimization (MOACO) algorithm. The implemented MOACO algorithm simultaneously minimizes the hop count and the number of disruptions, for a set of currently unicast working paths and a set of new unicast demands. The algorithm's output is a set of non-dominated solutions (known as Pareto set) to facilitate decision making. Simulation results prove the viability of MOACO algorithms and the advantage compared with the Integer Lineal Programming (ILP) approach.

Keywords: Optical Networks; Multi-Objective Optimization; Ant Colony Optimization

I. INTRODUCTION

Optical networks using wavelength-division multiplexing (WDM) technology are considered promising for transport networks, as they can satisfy the growing demand for bandwidth. The basic service in this kind of networks is a *lightpath*.

A lightpath is an optical communication path between two nodes. The optical links between intermediate Optical Cross-Connects (OXC) may have different wavelength assigned, or may be all at the same wavelength, according to the availability of conversion functions in the OXCs. In this work, we have considered the Wavelength Path (WP) network case, in which no wavelength conversion is allowed in the whole network and lightpaths are subject to the “wavelength continuity” constraint. Wavelength assignment to lightpaths in WP is an NP-complete problem, being equivalent to the well-known graph-coloring problem [1].

Typically, once the network is provisioned, the requests for new connections arrive dynamically over time. The lightpath establishment in this case is dynamic, and it is set up in real-time based on a routing and wavelength assignment (RWA) algorithm, without rearranging the working routes of existing lightpaths. As they reach their assigned period of life, they are retired, releasing their used links to be used for other requests. The network cannot afford to run optimization procedures to route every call that arrives dynamically. As a result of all these changes, the utilization of the network capacity slowly degrades to a point that calls may start getting blocked.

However, changes in core optical networks are made in the order of one each a few days, or even a few weeks, i.e. the traffic is pseudo-dynamic. In this scenario an optimization approach can be applied to get good network utilization performance. This is known as the *long-term reconfiguration problem* and it's an active area of research [19].

The long-term reconfiguration problem treats a static formulation of re-routing all working connections and optimizing the network capacity. However, it is possible that some of the current connections may be disrupted, which may not be acceptable. A reconfiguration algorithm should consider the trade-off between the optimality of the new lightpaths with regards to a given objective function, and the number of changes required from the current working paths which affects the amount of service disruption and switch re-tuning. This problem is known as *Minimal disruption-reconfiguration problem (MDR)*.

Several MDR approaches have been reported. Banerjee and Mukherjee in [12] and Ramamurthy and Ramakrishnan in [13] formulated the MDR using linear programming. Prathombutr [2] proposed a Multi-objective Evolutionary algorithm (MOEA), while Zheng et al. [14] proposed a Balanced Alternate Routing Algorithm (BARA) based on a genetic algorithm to solve the problem, but using a weighted combination of trade-off objectives when a (single objective) genetic algorithm is used.

Ant Colony Optimization (ACO) algorithms are inspired by the behavior of ant colonies searching for food. ACO has been established as a valid alternative in the resolution of hard

combinatorial optimization problems [3]. Particularly, ACO approach has been used to solve the RWA problem in recent works[20]. Therefore, in this work MOACO algorithms (Multi-objective ACO) were implemented for the MDR problem. The results from MOACO algorithms were compared to ILP results, to have a performance figure of the implemented algorithms. In our best knowledge the proposed approach has not been addressed yet.

The remainder of this work is organized as follows. A general definition of a Multi-objective Optimization Problem is presented in section 2. A mathematical problem formulation and the objective functions are given in section 3. The ACO and MOACO algorithms are explained in section 4. The experimental environment and results are given in section 5 and 6 respectively. Finally, conclusions and future works are presented.

II. MULTI-OBJECTIVE OPTIMIZATION

A general Multi-objective Optimization Problem (MOP) [18] includes a set of n decision variables, k objective functions, and m restrictions. Objective functions and restrictions are functions of decision variables. This can be expressed as:

Optimize

$$y = f(x) = (f_1(x), f_2(x), \dots, f_k(x)). \quad (1)$$

Subject to

$$e(x) = (e_1(x), e_2(x), \dots, e_m(x)) \geq 0, \quad (2)$$

Where the decision vector is:

$$x = (x_1, x_2, \dots, x_n) \in X$$

and the objective vector is:

$$y = (y_1, y_2, \dots, y_k) \in Y$$

X denotes the decision space while the objective space is denoted by Y . Depending on the kind of the problem, “optimize” could mean minimize or maximize. The set of restrictions $e(x) \geq 0$ determines the set of feasible solutions $X_f \subseteq X$ and its corresponding set of objective vectors $Y_f \subseteq Y$.

A multi-objective problem consists in finding the decision vector x that optimizes $f(x)$. In general, there is no unique “best” solution but a set of solutions, none of which can be considered better than the others when all k objectives are considered at the same time. This derives from the fact that there can be conflicting objectives, i.e., *trade-offs* between different objectives.

If a solution $u \in X_f$ is better than another one $v \in X_f$ in at least one of the k objectives and is not worst in any other objective, it is said that u dominates v denoted as $u \succ v$. If

neither v dominates u nor u dominates v , u and v are non-comparable, what is denoted as $u \sim v$. Alternatively, $u \triangleright v$ will denote that u dominates or is equal to v .

At the same time, a decision vector $x \in X_f$ is non-dominated with respect to a set $Q \subseteq X_f$ if and only if $x \triangleright q, \forall q \in Q$. If x is non-dominated with respect to the whole set X_f , it is called an optimal Pareto solution; therefore, the *Pareto optimal set* X_{true} may be formally defined as:

$$X_{true} = \{x \in X_f \mid \neg \exists y \succ x, \text{ where } y \in X_f\} \quad (3)$$

The corresponding set of objective vectors $Y_{true} = f(X_{true})$ constitutes the *Optimal Pareto Front*.

III. PROBLEM FORMULATION

An optical network is modeled as a directed graph $G = (N, L, W)$, where N is the set of nodes, L is the set of unidirectional links between nodes, with W wavelengths on each link.

Let be:

- $i \rightarrow s, d$ Index assigned to each source-destination pair of any demand request or existent connection, where s is the source node and d the destination node, with $s, d \in N$;
- p Index assigned to a lightpath for each i source-destination pair. A path has an associated wavelength $\lambda = 1, 2, \dots, W$. If k different routes are defined for a source-destination pair, then there are $p=1, 2, \dots, kW$ different lightpaths;
- $\psi_\lambda^{i,p}$ A binary wavelength indicator, which takes value of one if wavelength λ is used by the path (i, p) and zero otherwise (data);
- $\chi_{i,p}$ A binary path indicator which takes value of one if path (i, p) is a currently working path and zero otherwise (data – current traffic);
- $\varepsilon_l^{i,p}$ A binary link indicator, which takes value of one if link l is used in path (i, p) and zero otherwise, with $l \in L$ (data);
- $\delta^{i,p}$ A path indicator which takes value of one if lightpath (i, p) is chosen and zero otherwise (binary variable);
- c_i Demand for node-pair i , in terms of the number of requested lightpaths;

Using the above definitions, the MDR problem may be stated as a MOP searching the best solution that simultaneously minimize the following two objective functions:

1. *Hop count:*

$$y_a = \sum_i \sum_{p=1}^{kW} \delta^{i,p} \sum_{l=1}^L \varepsilon_l^{i,p} \quad (4)$$

2. *Number of disruptions in existing paths:*

$$y_b = \sum_i \sum_{p=1}^{kW} \chi^{i,p} (1 - \delta^{i,p}) \quad (5)$$

To solve this multi-objective problem using ILP, equation (5) is made equal to a constant and added to the known constraints, so the problem is converted in a Mono-objective one. With several runs, varying this constant, we find the Pareto front.

The optical layer constraints for our ILP formulation are:

A. *Link capacity constraint: Link l can connect only W lightpaths or less:*

$$\sum_i \sum_{p=1}^{kW} \delta^{i,p} \varepsilon_l^{i,p} \leq W \quad 1 \leq l \leq |L| \quad (6)$$

B. *Path wavelength usage constraint: Only one path can use a wavelength λ on link l :*

$$\sum_i \sum_{p=1}^{kW} \delta^{i,p} \varepsilon_l^{i,p} \psi_\lambda^{i,p} \leq 1 \quad 1 \leq l \leq |L| \quad (7)$$

C. *Demand constraint: Only c_i paths must be active for each node-pair.*

$$\sum_{p=1}^{kW} \delta^{i,p} = c_i \quad (8)$$

$$\forall i \rightarrow (s, d)$$

IV. MULTI-OBJECTIVE ANT COLONY OPTIMIZATION

Ant Colony Optimization (ACO) is a meta-heuristic inspired by the foraging behavior of ant colonies [4]. In the last few years, ACO has received increased attention by the scientific community as can be seen by the growing number of publications and the different fields of application [5]. Even though, there are several ACO variants. We next present what can be considered a standard approach [6].

A. *ACO Standard Approach.*

ACO makes use of simple agents called *ants* and a pheromone matrix $\tau = \{\tau_{jk}\}$ for iteratively constructing candidate solutions. Typically, the initial values are $\tau_{ik} = \tau_0 \forall j, k \in N$, where $\tau_0 > 0$. Furthermore, ACO algorithm takes advantage of problem dependent heuristic information using a parameter η_{ik} called *visibility*. The relative influence between the heuristic information and the pheromone levels is controlled using parameters α and β . While an ant is visiting node n , N_n represents the set of neighborhood nodes that are not yet visited from node n . The probability of choosing a node j while at node n , is defined by the following equation:

$$P_{nj} = \begin{cases} \frac{\tau_{nj}^\alpha \eta_{nj}^\beta}{\sum_{g \in N_n} \tau_{ng}^\alpha \eta_{ng}^\beta} & \text{if } j \in N_n \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

In order to avoid a premature convergence towards the local optimum, pheromones evaporation is applied for all node pairs (i) in the pheromone matrix according to $\tau_{jk} = (1 - \rho) \cdot \tau_{jk}$, where parameter $\rho \in (0, 1]$ determines the evaporation rate [4].

Let be U the new demands, CT current traffic and S a solution. Considering an elitist strategy, the best solution found so far S_{best} updates τ according to $\tau_{jk} = \tau_{jk} + \Delta\tau$, where typically $\Delta\tau(j, k) = 1/f(S_{best})$ if $(j, k) \in S_{best}$ and $\Delta\tau(j, k) = 0$ if $(j, k) \notin S_{best}$.

ACO is especially appealing when constructing solutions are needed; therefore, it seems interesting to study its application to the MDR Problem. Note that the standard approach optimizes a single objective. In the next section, this work presents a Multi-objective Ant Colony Optimization approach.

B. *Multi-Objective Ant Colony Optimization*

Alternative implementations of Multi-Objective Ant Colony Optimization (MOACO) algorithms derive from traditional ACO, based mainly on [8]:

- 1) the number of colonies with m -ants each
- 2) the way to manage the pheromone information, and
- 3) the heuristic functions.

Combinations of these characteristics are used to build MOACO algorithms.

In a mono-objective context, each ant that finds a new solution that is better than the best-found solution, so far, updates the pheromone matrix τ . Analogously, in a multi-objective context, the ants that build a good solution must also update Y_{known} , the set of best non-dominated solutions found by a running algorithm if and only if the new solution is non-dominated with respect to the whole Y_{known} found so far. At the same time, dominated solutions of the *Pareto Set* under construction must be eliminated. Pseudocode I present a general MOACO approach.

This work is based on the *Multi-Objective Ant Colony System* (MOACS) algorithm, which was demonstrated to have

good performance in several benchmarks. MOACS was proposed by Schaerer and Barán in [7]. It uses a pheromone matrix and two different visibilities, one for each objective of the problem, together with a pseudo-random rule presented in Pseudocode II to favor exploration. Every time an ant moves from state j to state n , it performs a local pheromone update. When a non-dominated solution is found, the Pareto set is updated and the algorithm resets the pheromone matrix, considering that the information was learned by means of sub-optimal dominated solutions [7]. This way, exploration is encouraged. If a recently found solution is non-dominated with respect to Y_{known} , a pheromone update is performed.

C. Ant-MDR:

MOACO algorithms are independent general-purpose methods of optimization. To particularize them to the MDR problem, an artificial ant should build the set of lightpaths that constitute a solution S . Such artificial ant for the MDR is denominated in this work *Ant-MDR*. Pseudocode III depicts the job of Ant-MDR.

Basically, an Ant-MDR takes randomly each unicast request from the set of demands $U \cup CT$, builds a path and assigns a wavelength, chosen randomly among all the available wavelengths common in every link of the path. If there is not any wavelength, a new path is build. Finally, a matrix that controls the available wavelengths in the network is updated.

For each node-pair i , the Ant-MDR must take the decision of disrupting the current lightpath for this request or looking for a new path. This decision must be taken at the beginning of the tour (when the current node is the destination node).

The probability of disrupting this request is:

$$pr_d^i = (\tau_d^i)^\alpha \cdot pr_0 \quad (10)$$

Where τ_d^i is a specific pheromone for the first objective (disruption) and pr_0 is an initial probability of disruption, very close to 0 for all the node-pair which belongs to CT, and 1 if not. The disruption pheromone τ_d^i is updated every time that an *colony* of m ants finish its trip, giving more chance to disruptions, if new solutions disrupts this request.

If the decision of disrupting was taken, then a new lightpath must be calculated using (9), and applying a new pair of pheromone τ and visibility η values, related to the second objective (hop count). The hop-count visibility η_{nj} is calculated from the well-known Dijkstra shortest-path algorithm, using net topology information and distance from the current node to the destination node. The hop-count pheromone τ_{nj} depends on how many times this link was used in the solutions of the Pareto front, for this specific request i .

PSEUDOCODE I: GENERAL MOACO APPROACH

Input: $\alpha, \beta, CT, U, \rho, \tau_0, m, G, pr_0, \eta$;

Output: Y_{true} ;

```

1:  $\tau = \text{Initialize\_Pheromones}(\tau_0)$ ;
2: do while stop criterion is not verified
3:    $t = t + 1$  /* generations
4:   for ant=1 to  $m$  /*  $m$  is the number of ants
5:      $S = \text{Ant\_MDR}(\alpha, \beta, CT, U, G, pr_0, \eta, \tau)$ ;
6:      $\text{Evaluate\_Solution}(CT, S)$ ; /* to calculate  $y_a$  and  $y_b$ 
7:      $Y_{true} = \text{Update\_Y}_{known}(S)$ ;
8:   end for
9:    $\tau = \text{UpdatePheromones}(Y_{true}, \rho)$ ;
10: end do
11: print  $Y_{true}$ ;
```

PSEUDOCODE II: THE PSEUDO-RANDOM PROPORTIONAL RULE

Input: q_0, n, N_n ;

Output: $j \in N_n$;

```

1: Select randomly  $q$ ; /*  $q, q_0 \in (0,1]$ 
2: if  $q > q_0$  then
3:   Choose node  $j$  with larger  $P_{jn}$ ; /* using equation (9)
4: else
5:   Randomly choose  $j$  using probability  $P_{jn}$ ;
6: end if
7: return  $j$ ;
```

V. EXPERIMENTAL ENVIRONMENT

Simulations were carried out using the backbone network topology illustrated in Fig. 1, with 14 nodes and 44 unidirectional links. It was derived by adding one span to NSFNET T1 backbone, to make each local degree more than two. Data regarding the physical topology of this network have been taken from [9].

We used two different sets of current (existent) paths for simulations:

1. The first one, shown in Table I, was derived from real traffic measurements which are reported in reference [9]. This matrix comprises 180 different paths. To simulate non-optimized use of the network, some shortest-paths were denied adding new constraints to the formulation, and solve an ILP optimization problem to have the current lightpath. We made $W=25$ (every unidirectional link has 25 available wavelengths), which is the minimum number to allocate all these routes.
2. The second set of current connections, with 250 source-destination pairs, was obtained by the procedure that follows:
 - A traffic generator chooses source-destination nodes i and time slot t_i associated to it.
 - An RWA algorithm (based on 3-disjoint shortest paths Dijkstra algorithm) assigns a lightpath to the request, if it is possible.

PSEUDOCODE III: THE ANT-MDR ALGORITHM

Input: $pr_0, \alpha, \beta, \tau_d, \tau, \eta, CT, U, G$;
Output: S ;

```

1:  $S = \emptyset$ ;  $success = true$ 
2: Initialize set  $Available = \{\text{all the lightpaths in the network}\}$ 
3: do while  $success = true$  or  $U \cup CT = \emptyset$  //all requests  $i$  processed
4:   choose a request  $i \in U \cup CT$ 
5:   if  $rand(0,1) < pr_d^i$  /* using (10)
6:     Verify the availability of  $lightpath_i^{CT} \in CT$  in  $Available$ 
7:     if  $lightpath_i^{CT} \in Available$ 
8:        $S = S \cup lightpath_i^{CT}$ 
9:        $disrupted = false$ 
10:       $success = true$ 
11:       $Available = Available - lightpath_{i,p}$ 
12:    endif
13:  endif
14:  if  $disrupted = true$  /* builds a new lightpath
15:     $lightpath_i^{NEW} = \{source_i\}$ 
16:    do while  $success = true$  or stop criterion is reached
17:       $n = \text{source node of } i$ 
18:      do while the destination node for  $i$  is not reached
19:        Build the set of feasible nodes  $F_n$ 
20:        Assign probability  $pr_{nj}$  to each node of  $N_n$ ;
21:        Select randomly node  $j$  of  $F_n$  /* by Pseudocode 2
22:         $lightpath_i^{NEW} = lightpath_i^{NEW} \cup j$ 
23:         $n = j$ 
24:      end do
25:      Build the set  $W_i$  of available wavelengths for  $lightpath_i^{NEW}$ 
26:      if  $W_i = \emptyset$ 
27:         $success = false$ 
28:      else
29:        Assign randomly a wavelength  $\gamma_i \in W_i$  to  $lightpath_i^{NEW}$ 
30:         $success = true$ 
31:         $S = S \cup lightpath_i^{NEW}$ 
32:         $Available = Available - lightpath_i^{NEW}$ 
33:      endif
34:    end do
35:  end if
36:   $U \cup CT = U \cup CT - (i, lightpath_i)$ 
37: end do
38: if  $success = true$ 
39:   return  $S$ 
40: else
41:   Request  $i$  not satisfy, return Error
42: end if

```

- A traffic simulator deletes lightpaths that reached their time of operation, and new routes are added to the network.

The final demand matrix was obtained adding a set of new requirements to the current connections. In this way, adding two different sets of new demands to each set of existent connections, we had four multi-objective problems, (instances A, B, C and D), supposing fixed the current working paths.

To obtain the ILP results, we keep fixed one of the objectives (number of disrupts) and treat it as a mono-objective problem. Several runs, varying the quantity of disrupts allowed, give us the points of the Pareto front. The software tool GAMS 22.8 based on branch and bound method was used. ILP was limited to the three disjoint shortest paths for each unicast request.

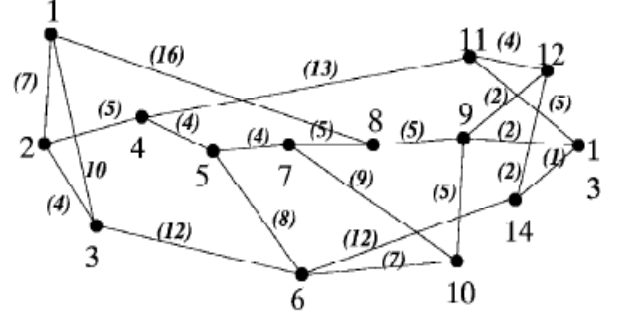


Figure 1. NSFNET T1 network backbone, with 14 nodes and 44 unidirectional links

TABLE I. CURRENT CONNECTIONS MATRIX

	2	3	4	5	6	7	8	9	10	11	12	13	14
1	1	0	0	0	0	0	0	1	0	0	0	0	0
2		1	0	7	1	2	6	2	4	0	4	0	3
3			1	2	0	10	21	4	1	0	0	0	1
4				0	0	1	0	1	1	0	1	0	1
5					0	2	6	2	2	0	8	1	4
6						1	0	1	1	0	0	0	1
7							11	3	2	0	7	0	3
8								6	6	1	5	1	3
9									4	1	13	2	0
10										1	2	1	0
11											1	0	0
12												0	0
13													1

Numbers of source-destination nodes are related to Fig.1

MOACO algorithms have been implemented on a AMD Turion64 x2 computer with 3GB of RAM with a gcc compiler.

For these experiments, the results of the proposed MOACOs have been compared to the ILP results.

With the objective of evaluating the proposed methods, many simulations were performed with each unicast group. In that context, the used parameters were: 40 ants for each colony, $\beta/\alpha=4$ relative importance, evaporation percent $\rho = 0.10$, a pseudo-random probability $q_0= 0.95$ a disruption probability $\rho_0=0.01$, the stopping criterion was 50000 iterations.

For each unicast group U , a set of approximate solutions to the Pareto Front was calculated using the following procedure:

1. The MOACO algorithm was run 10 times.
2. Ten set of non-dominated solutions: $Y_1, Y_2 \dots Y_{10}$, were calculated, one for each run.
3. Dominated solutions were deleted, and an approximation set of the *Pareto Front*, called " Y_{known} ", is created.
4. Then, each of the 10 runs was compared with Y_{known} to calculate an average of the Pareto Metrics.

VI. EXPERIMENTAL RESULTS.

This section shows a comparison between the solutions found with the implemented MOACO and ILP algorithms with respect to Y_{known} .

Fig. 2, 3, 4 and 5 present the Pareto fronts for each combination of two sets of existent connections and two sets of new demands.

To compare the obtained results of ILP and MOACO we used two metrics that help evaluate them considering different quality criteria. Let be PF and PF' the sets of proposed solutions of two different algorithms. The used metrics are:

- a. *Coverage C (PF/PF')*: Defined in [15] as the relation between the quantity of dominated solutions of PF' and the total quantity of solutions in PF'.
- b. *Overall Non-dominated Vector Generation ONVG*: The total number of solutions [16].

The results can be seen in Table II. Both metrics are shown with normalized values, for the 4 different instances. The best algorithm, in a given criteria, shows a value of 1 in this table.

Notice that, in every instance MOACO algorithms give results very close to ILP, and even get better results in areas of the Pareto Front with low quantity of disruptions, which can be very interesting for the network administrator. As an example, see results for Instance D (Figure 5). ILP gives solutions with at least 4 disruptions, while MOACO shows alternatives with 0, 1, 2 or 3 disruptions, with less QoS loss.

Also notice that in every instance MOACO algorithms found at least one non-dominated solution that didn't find ILP algorithm, which is supposed to be an exact method (in this case explores 3 disjoint shortest paths per source-destination pair). MOACO algorithms, with the above restriction, show more power of exploration for disruptions in the range of 0-8. ILP is NP-hard and therefore less scalable to more complex topologies, in which this technique is impractical to apply.

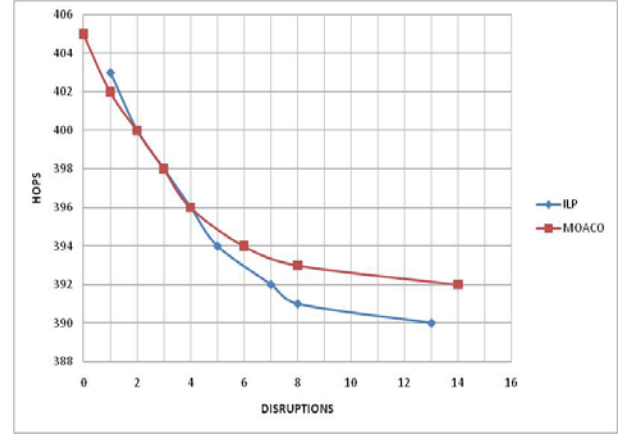


Figure 2. MOACO and ILP Pareto front for Instance A. Existent connections from Table II, 2 new added requests to the current connections.

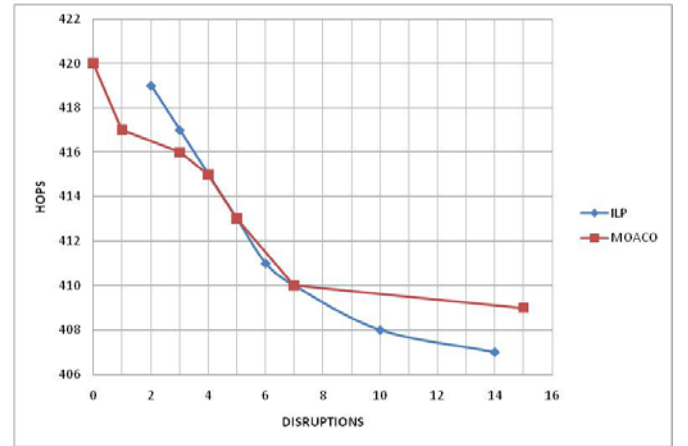


Figure 3. MOACO and ILP Pareto front, for Instance B. Existent connections from Table II, 8 new added requests to the current connections.

As can be seen in Table II, ILP outperformed MOACO in general results. However, for low number of disruptions, MOACO found solutions that ILP didn't, which is very important for QoS reasons. Besides, the close approximation of the MOACO solutions to the ILP is promising, considering the high exploration capacity of MOACO, while ILP is restricted to the three disjoint shortest paths for each pair of source-destination node.

VII. CONCLUSIONS

We implemented MOACO algorithms to solve the Minimal Disruption-Rerouting (MDR) problem, optimizing simultaneously two objective functions: (1) the hop count and, (2) the number of disruptions, subject to the wavelength continuous constraint. We used two different working paths, the

first one showing a real distribution of requirements over NSF network, obtained from [9], and a second one obtained with a traffic generator, with uniform distribution. Two different sets of new demands were added to these current connections, giving four different instances. A novel MOACO algorithm was developed, adapting it to MDR problem (*Ant-MDR*). These algorithms calculate a whole set of optimal Pareto solutions in one run. This last feature is especially important since the decision maker can choose from the most adequate solution set for each particular case, without *a priori* restrictions that could eliminate good compromise solutions. Typically this occurs in the multi-objective optimization approach with weighted sum or lexicographic order.

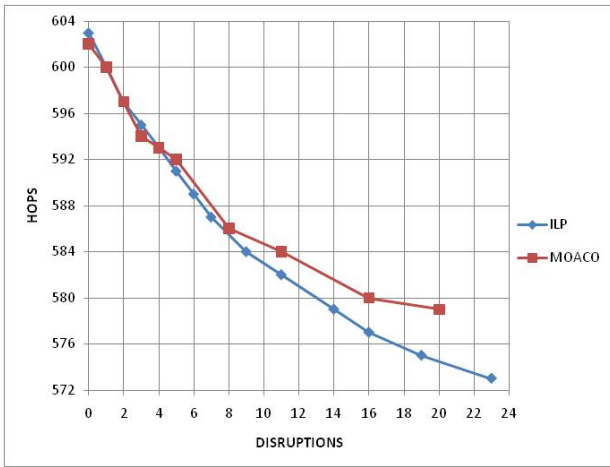


Figure 4. MOACO and ILP Pareto front for Instance C. 8 new added requests to current connections obtained with a traffic simulator.

To validate the proposed MOACO approach, an ILP formulation was also implemented for comparison reasons. After an extensive simulation testing partially presented in this work, we obtained four graphics showing the results of both methods.

Considering the preliminary results, we can conclude that the proposed approach is promising to get good results with respect to the exact approach (ILP). However, it is clear the need to improve the convergence towards the real Pareto Front.

As future work, the authors plan to add heuristics to the *ant-MDR* algorithm that help finding good solutions for high number of disruptions. One possible heuristic is solving the problem in specific areas of the Pareto Front, keeping control of number of disruptions that are being calculating. Also, they intend to extend testing over larger network topologies, considering new objective functions as the *reliability* and using other multi-

objective metrics. Besides, the calculation of backup paths for different levels of service may be added to the formulation.

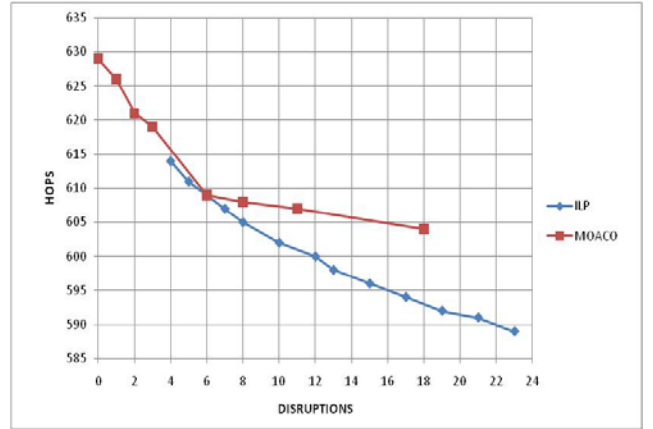


Figure 5. MOACO and ILP Pareto front for Instance D, 13 new added requests to the current connections.

TABLE II. COMPARISON BETWEEN MOACO AND ILP RESULTS

	COVERAGE	ONVR	PERFORMANCE
INSTANCE A			
MOACO	0,333	1	0,667
ILP	1	1	1
INSTANCE B			
MOACO	1	0,875	0,938
ILP	0,500	1	0,750
INSTANCE C			
MOACO	0,556	0,833	0,694
ILP	1	1	1,0
INSTANCE D			
MOACO	0,375	0,833	0,604
ILP	1	1	1,0
GENERAL PERFORMANCE			
MOACO			0,726
ILP			0,938

REFERENCES

- [1] I. Chlamtac, A. Ganz and G. Karni. "Lightpath Communications: An Approach to High Bandwidth Optical WANs". IEEE Transactions on Communications. Vol. 40, No. 7, pp. 1171-1182, July, 1992.
- [2] P.Prathombutr. "Virtual Topology Reconfiguration in wavelength-routed optical networks". Doctoral Thesis, University of Missouri-Kansas City,2003.
- [3] M. Dorigo, V. Maniezzo and A. Colomi. "The Ant System: Optimization by a colony of cooperating agents". IEEE Transactions on Systems, Man, and Cybernetics-Part B, Vol. 26, No.1, pp. 1-13, 1996.

- [4] M. Dorigo and G. Di Caro. "The Ant Colony Optimization meta-heuristic". In *New Ideas in Optimization*. McGraw Hill, London, UK, 1999.
- [5] T. Stützle and H. Hoos. "Max-Min Ant System". *Future Generation Computer System* 16(8): pp. 889-914, June, 2000
- [6] M. Guntsch and M. Middendorf. "A Population Based Approach for ACO". In Stefano Cagnoni, Jens Gottlieb, Emma Hart, Martin Middendorf, and Günther Raidl, *Applications of Evolutionary Computing, Proceedings of EvoWorkshops2002: EvoCOP, EvoLASP, EvoSTim*, Springer-Verlag, vol. 2279, pp. 71-80, Kinsale, Ireland, 2002.
- [7] M. Schaerer and B. Barán. "A multiobjective Ant Colony System for Vehicle Routing Problems with Time Windows" *Proc. Twenty first IASTED International Conference on Applied Informatics*, Innsbruck, Austria, pp. 97-102, 2003.
- [8] M. Lopez-Ibañez, L. Paquete and T. Stützle. "On the design of ACO for the Biobjective Quadratic Assignment Problem". In: Dorigo, et al. (Eds.): *Proc. of the Fourth International Workshop on Ant Colony Optimization (ANTS 2004)*, Lecture Notes in Computer Science, Springer Verlag.
- [9] Y. Miyao and H. Saito. "Optimal Design and Evaluation of Survivable WDM Transport Networks". *IEEE Journal on selected areas in Communications*, Vol.16, No 7, September 1998, pp 1190-1198
- [10] H. Zang, R. Huang, and J. Pan. "Methodologies on designing a hybrid shared-mesh protected WDM network with sparse wavelength conversion and regeneration". *Proc. of APOC 2002*, Shanghai, China, pp. 188-196, 14-18 October, 2002.
- [11] T. Li and B. Wang. "Cost effective shared path protection for WDM optical mesh networks with partial wavelength conversion". *Photonic Network Communications*, vol. 8, no. 3, pp. 251-266, November, 2004.
- [12] D. Banerjee and B. Mukherjee, Wavelength-routed Optical Networks: Linear Formulation, Resource Budget Tradeoffs, and a Reconfiguration Study, *IEEE/ACM Transactions on Networking*, Vol. 8, No. 5, pp. 598-607, October 2000.
- [13] B. Ramamurthy and A. Ramakrishnan, Virtual Topology Reconfiguration of Wavelength-Routed Optical WDM Networks, *Proc. of Global Telecommunications Conference*, Vol. 2, pp. 1269-1275, 2000.
- [14] J. Zheng, B. Zhou and H. T. Mouftah, "Design and Reconfiguration of Virtual Private Networks (VPN) over All-Optical WDM Networks", *Proc of 11th. International Conference on Computer Communications and Networks*, pp 599-602, October 2002.
- [15] E. Zitzler, K. Deb, y L. Thiele. Comparison of Multiobjective Evolutionary Algorithms: Empirical Results. *Evolutionary Computation*, 8(2):173-195, Summer 2000.
- [16] D. A. van Veldhuizen. *Multiobjective Evolutionary Algorithms: Classifications, Analyses, and New Innovations*. PhD thesis, Department of Electrical and Computer Engineering, Graduate School of Engineering, Air Force Institute of Technology, Wright-Patterson AFB, Ohio, Mayo1999.
- [17] N. M. Duarte, A. E. Ruano, C. M. Fonseca, y P. J. Fleming. Accelerating Multi-Objective 187 Control System Design Using a Neuro-Genetic Approach. In *2000 Congress on Evolutionary Computation*, volume 1, págs. 392-397, Piscataway, New Jersey, Julio 2000. IEEE Service Center.
- [18] G. Varela and M. Sinclair. "Ant Colony Optimization for Virtual-Wavelength-Path Routing and Wavelength Allocation". *Proc. Congress on Evolutionary Computation (CEC'99)*, Washington DC, USA, pp. 1809-1816, July, 1999
- [19] Arun K. Somani. "Survivability and Traffic Grooming in WDM Optical Networks" ISBN 0511137540 – Cambridge University Press, 2005.
- [20] A. Arteta, B. Baran, D. Pinto. "Routing and Wavelength assignment over WDM optical networks. A Comparison between MOACOs and classical approaches" III IFIP/ACM Latin American Networking Conference, Costa Rica, Oct 2007.